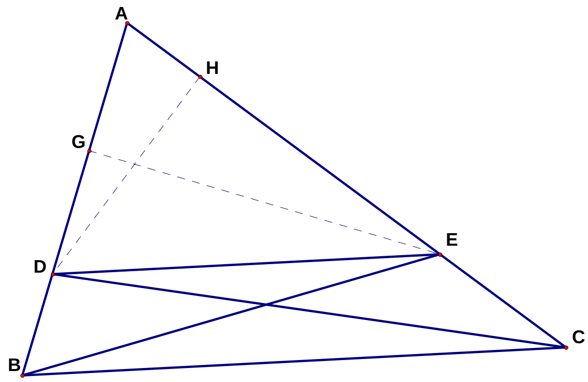
Proportionality in a Triangle Theorem



Theorem: **If a line intersects two sides of a triangle, it is parallel to the third side if and only if the segments created by the intersection points are proportional.**

Part 1

Given: $\triangle ABC$ with point D on $\overline{AB}$ and point E on $\overline{AC}$ such that $\overline{DE}$ is parallel to $\overline{BC}$

Prove: $\dfrac{AD}{DB} = \dfrac{AE}{EC}$

Note: I have added points G and H such that EG is perpendicular to AB and DH is perpendicular to AC.

1. Prove that the (area of $\triangle BDE$) $=$ (area of $\triangle CED$)
2. Prove that the (area of $\triangle BDC$) $=$ (area of $\triangle CEB$)
3. Prove that the (area of $\triangle ADC$) $=$ (area of $\triangle AEB$)
4. Use your above results to show that $\dfrac{AB \cdot GE}{AD \cdot GE} = \dfrac{AC \cdot DH}{AE \cdot DH}$ or alternatively (and maybe easier), show that $\dfrac{AB \cdot GE}{DB \cdot GE} = \dfrac{AC \cdot DH}{EC \cdot DH}$
5. Use this to show that $\dfrac{AD}{DB} = \dfrac{AE}{EC}$

Part 2

Given: $\triangle ABC$ with point D on $\overline{AB}$ and point E on $\overline{AC}$ such that $\dfrac{AD}{DB} = \dfrac{AE}{EC}$

Prove: $\overline{DE}$ is parallel to $\overline{BC}$

Definition

Two polygons are *similar* if and only if their corresponding angles are equal and the corresponding sides are proportional.